When Does Generative Augmentation Help Extreme Coastal Water-Level Forecasting? Architecture Dependence, Physical Priors, and Independent-Event Validation

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Preprint — evidence-linked; hypotheses not yet confirmed

Abstract

Extreme coastal water levels are rare, and generative augmentation is the obvious way to make them less rare in training data. We ask when it actually helps, and report a programme of controlled experiments in which the honest answer is: not yet demonstrably, and for reasons that can be stated precisely. Across five regions under a frozen 96 h→72 h contract (2015–2024), we first reconstruct the independent-event census from raw cluster artefacts. The 2015–2024 union is **679** event units (634 peak-over-threshold clusters, 73 tropical cyclones, 28 matched), not the 707 previously reported; the 2024 test split contains 59 independent units, with no duplicate timestamps and no cross-split window leakage. We then benchmark four conditional generators on identical splits. Only Flow Matching reproduces the target dispersion (generated/real scale ratio 0.945–1.116, Wasserstein-1 0.047); conditional VAE and TimeGrad are under-dispersed by factors of 2.3 and 12, and Diffusion-TS is rejected in all ten runs. An oracle-denoiser test proves the reverse diffusion recursion itself correct (max absolute reconstruction error 1.1×10^{-10}), which localises the failures to implementation defects rather than model families; we report that two of the three proposed repairs did *not* work. Finally, after correcting cross-architecture pairing, a 5,000× p-value inflation and a Monte-Carlo resolution floor that made significance arithmetically unreachable, the corrected event-level analysis finds **zero seed-robust augmentation findings** across 320 comparisons. Effects are architecture-dependent (the TCN responds roughly forty times more strongly than the GRU) and region-dependent (Thames is harmed for both), and baseline seed variance alone reaches 25%. We release the verified event census, the fidelity benchmark, the corrected statistics and a three-year split protocol built from scratch, and we state what would be required to establish a positive claim.

1 Introduction

Coastal flooding is among the most costly natural hazards, and the events that matter most operationally are the ones that are rarest in the historical record. A forecaster trained on ten years of gauge observations therefore spends most of its capacity on benign conditions. Generative augmentation is the obvious response: synthesise additional extreme residuals so that the downstream forecaster sees more of the tail. The question this paper asks is narrower and more useful than "does augmentation work": *when* does generative augmentation help extreme coastal water-level forecasting, and what does the answer depend on?

We began with the standard hypothesis, stated here in its original form so that the reader can see what the evidence did to it:

H1: Tail-aware conditional generative augmentation, applied under a matched forecaster training budget, improves prediction accuracy on *independent held-out real storm events* relative to real-only training, standard oversampling, perturbation augmentation, and standard conditional diffusion.

H1 is still not established. What this paper contributes is the reason it is not established, and the conditions under which a future study could establish it.

1.1 Why the question is harder than it looks

Three problems recur in published and internal evaluations of generative augmentation for time series, and all three were present in our own earlier work.

1. **Window overlap masquerades as sample size.** A 96 h→72 h contract produces tens of thousands of sliding windows per region, but a single storm supplies hundreds of them. Treating windows as independent inflates every confidence interval by roughly the square root of the within-event window count. The unit of inference must be the *event*, not the window.

2. **Event counts are easy to double-count.** Tropical-cyclone catalogues and peak-over-threshold (POT) declustering describe overlapping phenomena. Adding the two totals without computing the intersection inflates the census; in our own repository this produced a decade count of 707 instead of 679 (Section 6.1).

3. **Generator failures are attributed to families rather than to code.** A diffusion model whose reverse sampler is correct but whose Fourier regulariser regresses the wrong target will fail, and it is tempting to conclude that diffusion is unsuited to the problem. It is not a family-level conclusion; it is a line-level defect (Section 6.3).

1.2 Contributions

1. **A verified event census.** We recompute the 2015–2024 independent-event union for five regions from raw cluster artefacts (634 POT clusters, 73 tropical cyclones, 28 matched ⇒ **679** units), and show that the 2024 test split contains **59** independent units. Duplicate timestamps and cross-split forecast-window leakage are checked and found absent.

2. **A four-generator benchmark on identical splits.** Conditional VAE, TimeGrad, Diffusion-TS and Flow Matching are compared under one contract with one fidelity gate. Flow Matching is the only generator that reproduces the target dispersion (0.945–1.116), and even it does not clear the full gate.

3. **A diagnostic recovery study.** We prove the DDPM reverse recursion correct with an oracle denoiser (max absolute error 1.1×10^{-10}), localise three implementation defects, and report honestly that *two of the three proposed repairs did not work* — the repaired VAE samples *worse*, and the repaired Diffusion-TS diverges to NaN. This is a negative result about our repairs, not about the families.

4. **A corrected event-level evaluation.** After fixing cross-architecture pairing, a 5,000× p-value inflation and a Monte-Carlo resolution floor above the correction threshold, we find **zero** seed-robust augmentation findings across 320 comparisons. Effects are architecture-dependent (TCN responds, GRU barely does) and region-dependent (Thames is harmed), and baseline seed variance alone reaches 25%.

5. **A physics-prior pilot and a three-year protocol.** Causal ERA5-conditioned Flow Matching variants are trained and evaluated (Section 6.6), and a 2015–2020/2021/2022–2024 protocol is implemented from scratch because no existing checkpoint can be evaluated on 2022–2024 (Section 6.7).

1.3 Paper structure

Section 2 positions the work. Section 3 describes the frozen data protocol. Section 4 describes the generators and the forecasters. Section 5 states the experimental design. Section 6 reports the census, the generator benchmark, the recovery study, the corrected event-level analysis, the physics pilot and the expanded protocol. Section 7 explains the mechanisms; Section 8 states what would be required to move H1 from untested to established.

2 Related Work

2.1 Coastal water-level forecasting

Traditional coastal water-level forecasting combines deterministic tidal constituents with empirical or dynamical storm-surge models [Yang et al., 2023]. In recent years, neural approaches have improved forecast quality on held-out gauges. Sergeev et al. [2024] demonstrated that information from multiple gauges along a coastline substantially improves forecast accuracy for downstream stations. Tong et al. [2024] introduced dual-timescale residual memory that separately models slow tidal evolution and fast storm-driven residuals, with particular attention to extreme events. These works establish that (a) the de-tided residual signal carries the most operationally relevant information during storms, and (b) extreme events demand model architectures that differ from those optimized for benign conditions.

2.2 Data augmentation for time-series forecasting

Data augmentation for time series is reviewed by Hoppe et al. [2024] and related surveys. Standard approaches include: (1) *oversampling* rare events or weighting them in the loss function, (2) *perturbation* with additive or multiplicative noise, (3) *interpolation* between adjacent real windows, and (4) *generative models* including GANs, VAEs, and diffusion. A key distinction for coastal water levels is whether augmentation preserves the *physical plausibility* of the residual signal: unrealistic samples may actually degrade downstream forecasting by corrupting the training distribution.

2.3 Conditional diffusion models for time series

Conditional score-based diffusion models have been applied to time-series imputation [Tashiro et al., 2021, Ma et al., 2023] and generation. The canonical formulation [Ho et al., 2020] defines a forward process $q(x_t|x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)I)$ and trains a network $\epsilon_\theta(x_t, t, c)$ to predict the injected noise, where c denotes conditioning information. During generation, the reverse process iteratively denoises from T back to 0:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t, c)\right) + \sigma_t z, \tag{1}$$

where $\sigma_t^2 = \beta_t(1-\bar{\alpha}_{t-1})/(1-\bar{\alpha}_t)$ is the posterior variance and $z \sim \mathcal{N}(0, I)$.

2.4 This work's niche

To our knowledge, this is the first systematic application of conditional diffusion augmentation to coastal water-level forecasting with (a) event-grouped leakage-safe splits, (b) train-only threshold estimation, (c) matched-budget comparison across augmentation methods, and (d) pre-registration of hypotheses before running experiments.

3 Research Protocol

3.1 Data source and admission

The authorized owner copy of the dataset is at `/data/yingda/tidetail-data/prime-formal-2015-2024`, containing 350 files totaling 451 MB. Six coastal regions are included, with 87,672 hourly UTC rows each spanning 2015-01-01 to 2024-12-31. Region admission status is recorded in `region_availability.json`:

Table 1: Region formal training authorization status

Region	Training authorized
New York Bight	Yes
Thames / North Sea	Yes
Hong Kong Pearl River	**No** (license/datum not signed)
Southeast Australia	Yes
US Gulf Coast	Yes
Charleston Atlantic	Yes

The Hong Kong region is excluded from all formal experiments until the data owner confirms license, vertical datum, and the HK event `pearl_river_hong_kong_event_01149`, which spans the validation/test boundary at index 78904. The migration manifest (`MIGRATION_MANIFEST.json`) exists one level above the release directory; file SHA256 and size checks confirm all inspected files match both release and migration inventory entries.

3.2 Target definition

The release manifest `database_manifest.json` defines:

- **Total water level**: observed water level above local chart datum.
- **Tide**: deterministic FES2022 tidal constituent forecast (no future uncertainty in this archival release).
- **Residual**: water_level_m = observed_total − FES2022_tide.

The pilot diffusion model targets `water_level_m` (residual), in metres, with a fixed 72-hour history → 24-hour forecast window for engineering feasibility. The formal release contract specifies 96-hour history → 72-hour forecast; all comparisons between methods should use one consistent contract for scientific validity.

3.3 Leakage-safe splitting

Our protocol enforces three hard gates before any window creation or normalization:

1. **Event-group disjointness**: the same independent storm event may not appear in two partitions. Events are represented as `EventGroup(event_id, start_idx, end_idx)` intervals. The implementation in `src/tidetail/splits.py` validates non-overlapping intervals and unique IDs before splitting.

2. **Chronological ordering**: train row block must end before validation row block, which must end before test row block. The split is 8:1:1 by chronological event-group order, subject to `min_events_per_split`.

3. **Full-support embargo**: for forecast origin time o, lookback L, and horizon H, the full temporal support of one forecast window is $[o - L, o + H)$. Two partitions must not have overlapping full-support intervals. The repaired `embargo_filter` compares every index in an event group to every protected index, replacing the earlier midpoint-only comparison that incorrectly accepted dangerous events (see Appendix A). Additionally, `embargo_full_support_filter` enforces $o \geq$ split_block_start $+ L$ and $o \leq$ split_block_end $- H$ for each forecast origin.

All scaler fitting, quantile threshold estimation, and interpolation fitting are performed on *train rows only* and then applied to all partitions without re-fitting. Threshold-run events crossing the train boundary are removed in full before window creation, not partial-interval trimmed.

3.4 Formal release contract

The database manifest specifies:

- **Input**: 96 hours of historical observations.
- **Horizon**: 72 hours of forecast.
- **Split**: chronological 8:1:1 with strict window containment.
- **Tide**: existing local FES2022 deterministic archive.
- **Forcing**: ERA5 hindcast only. No operational 72-hour forecast claim without archived issue/valid/lead NWP forecast vintages.

The ERA5 forcing data available in this release is historical hindcast data, not forecast data. Using future observed ERA5 values as conditioning information in a forecast experiment would constitute leakage unless explicitly labelled as an oracle experiment. Our gauge-only diffusion model does not consume atmospheric forcing; this limitation is discussed in Section 7.

4 Method

4.1 Conditional DDPM architecture

We implement a 1D temporal denoising diffusion probabilistic model with the following components:

1. **Temporal embedding**: sinusoidal frequencies $f_j = 10000^{-2j/D}$ embedded into a width-`width` MLP, where D is half the width.

2. **History conditioning**: a separate MLP consumes all 96 past observations (72 in the pilot), producing a `width`-dimensional conditioning vector that is *added* to the temporal embedding before entering the denoiser.

3. **Dilated residual blocks**: four blocks with dilation $\{1, 2, 4, 8\}$, each containing GroupNorm (groups=4), SiLU activation, 3-conv padding to preserve length, and a linear condition injection.

4. **Output head**: GroupNorm + SiLU + 3-conv producing a single-channel epsilon prediction at each of the 24 (pilot) or 72 (main-task) forecast horizons.

Total parameter count: 27,169 (pilot 72→24). For the main task 96→72, the horizon changes to 72 and the conditioning input to 96 values; parameter count remains in the 30–40K range, matched to the GRU and TCN budgets below.

4.2 Noise schedule and forward process

We implement cosine cumulative noise schedule [Ho et al., 2020] with $T = 64$ diffusion steps in the pilot:

$$\beta_t = 1 - \frac{\cos((t + 0.008)/1.008 \cdot \pi/2)^2}{\cos((t - 1 + 0.008)/1.008 \cdot \pi/2)^2}, \qquad \alpha_t = 1 - \beta_t, \qquad \bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i. \tag{2}$$

At $t = T = 64$, the historical schedule reaches $\beta_{64} \approx 0.999$ and $\bar{\alpha}_{64} \approx 5.93 \times 10^{-7}$, producing $\approx 1299\times$ clean amplification at the final reverse step. An oracle comparison (Section 6) shows that subsampling a $T' = 1000$ cosine schedule to 64 steps reaches $\bar{\alpha}_{64} \approx 5.89 \times 10^{-4}$ (41× amplification), which eliminates the catastrophic scale explosion.

4.3 Training objective

Training targets *unit Gaussian epsilon* at each timestep. The loss is mean squared error between the network's epsilon prediction and the actual injected Gaussian noise:

$$\mathcal{L} = \mathbb{E}_{t, x_0, \epsilon \sim \mathcal{N}(0,I)} \left[\left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, t, \text{history}) \right\|^2 \right]. \tag{3}$$

Training hyperparameters for the pilot run (Issue #6): AdamW with lr = 0.001, batch size 128, three epochs, BF16 precision on RTX3090 GPU 4, seed 20261025.

4.4 Forecasting baselines

All baselines consume 96 hours of hourly history and produce 72 hours of forecast, using the same train-only normalization as the diffusion model. Missing-value masks are concatenated as an extra input channel.

Rule-based baselines. *Persistence* carries forward the last observed water level for all 72 hours. *Seasonal baseline* repeats the most recent semi-diurnal tidal cycle (144 hours ≈ 6 days) to capture the dominant tidal periodicity.

GRU encoder-decoder. A 2-layer GRU encoder maps the 96-hour history into a hidden state; a GRU decoder then autoregressively produces 72 forecast steps, each step feeding its own output back as input. Hidden size is matched to DDPM parameter budget ($\approx$ 5–10K). The GRU operates on normalized values with explicit gradient tracking for training and a no-grad `predict_eval` path for inference.

Temporal Convolutional Network. A stack of dilated causal 1D convolutions with Chomp1d padding ensures causality. Dilation grows exponentially ($\{1, 2, 4, 8\}$) to achieve a receptive field covering the full 96-hour history. The final hidden-state channel at the latest timestep is projected to all 72 horizons *at once* (non-autoregressive). Channel count is tuned to match DDPM budget ($\approx$ 15–20K).

4.5 Training-set augmentation

All augmentations modify the training set only. Validation and test data are never augmented. Three augmentation strategies are implemented:

1. **A1 — Balanced extreme-event oversampling.** Windows where any target cell exceeds the train-only q95 threshold receive a higher sampling weight ($\approx 3\times$) in each optimizer step. This addresses the rare-event training imbalance without fabricating new observations. Implemented via `torch.utils.data.WeightedRandomSampler`.

2. **A2 — Residual noise perturbation.** Training history (not target) is perturbed with additive Gaussian noise whose standard deviation is 30% of the train-only residual standard deviation. This models instrument uncertainty and sub-hourly unresolved variability. Targets are *never* modified.

3. **A2 — Tidal phase shifting.** Both history and target windows are circular-shifted by a random fraction of the semi-diurnal tidal cycle (12-hour period, maximum shift 6 hours). This exploits tidal phase symmetry to create valid phase-shifted training pairs while preserving the residual forecast relationship.

4.6 Engineering validation guards

After the pilot revealed catastrophic scale explosion, we added validation guards that *reject* nonfinite or implausibly scaled samples. These are engineering checks, not model repairs:

1. **Nonfinite rejection**: raise on nonfinite conditioning history, nonfinite epsilon prediction at any reverse step, or nonfinite reverse sample intermediate state.

2. **Train-reference scale screen**: after de-normalization, compare sample standard deviation to frozen training target standard deviation. Reject if ratio < 0.1 (collapse) or > 5.0 (dispersion), or if mean offset exceeds 5 training standard deviations. Both bounds are explicit policy defaults, not estimates of physical plausibility.

The historical GPU pilot predates these guards; its samples have SD ratio 22.95 and would be rejected by the screen.

5 Experimental Setup

5.1 Frozen forecast contract

The main task is **96 hours of hourly history → 72 hours of future forecast**. The target is `water_level_m`, which is *already* the non-tidal residual: the database satisfies the exact identity `water_level_m` + `tide_level_m` = `observed_total_water_level_m` to machine precision (MAE = 0.00000000 m, $r = 1.000000$, verified by `tests/test_target_semantics_contract.py`). The residual must therefore never be de-tided a second time; an earlier pipeline that computed `water_level_m` − `tide_level_m` injected a spurious oscillatory component and its checkpoints are discarded. The historical pilot (Issue #6) used 72h→24h on de-tided residuals only; that model is classified as exploratory and *must not* be compared directly against main-task metrics.

Five regions are admitted: `ny_bight`, `thames_north_sea`, `southeast_australia`, `us_gulf_coast`, `south_atlantic_charleston`. Hong Kong (`pearl_river_hong_kong`) is excluded pending license/datum/signoff.

5.2 Split protocol

- **Event-grouped chronological 8:1:1 split**. Each partition contains independent storm events in chronological order, with no event appearing in two partitions.
- **Full-support embargo**: a forecast origin o is safe only if $o \geq a + 96$ and $o \leq b - 72$, where $[a, b)$ is the partition's row block. This ensures no window's support $[o - L, o + H)$ crosses a partition boundary.
- **Train-only scalers**: mean/SD fit on train rows only, applied without re-fit to all partitions.
- **Train-only thresholds**: q95/q99 from train rows only, used for tail metrics and oversampling weights.
- **Missing masks**: all models consume a boolean mask per hour, fixed and common across all A0–A4 methods.

5.3 Evaluation metrics

All metrics are organized by their statistical unit (hourly cell, threshold event, or independent storm event), as defined in the companion document `docs/evaluation/METRIC_DEFINITIONS.md`:

All threshold denominators are explicit (zero-denominator returns NaN, never an implicit zero). The FPR definition follows standard clinical-statistics usage: FP / (FP + TN), not FP / (TP + FP) (the latter is FAR).

5.4 Comparison groups

The five A0–A4 groups share identical downstream predictor architecture, optimizer budget, training sample budget, and random seeds. Augmentation fraction is matched across A3/A4.

- **A0**: Real-only training, no augmentation.
- **A1**: Balanced extreme-event oversampling via WeightedRandomSampler.
- **A2**: Residual-noise + tidal-phase-shift augmentation on training set only.

Table 2: Evaluation metrics by statistical unit

Unit	Metric	Definition
Hourly cell	MAE / RMSE	Absolute error over finite cells
	Tail MAE / RMSE (q95)	Cells where $y \geq q95_{\text{train}}$ only
	Tail MAE / RMSE (q99)	Cells where $y \geq q99_{\text{train}}$ only
	Lead-time MAE / RMSE	Per horizon hour (1–72)
	Station-wise MAE / RMSE	Per gauge station
Threshold event	Precision	TP / (TP + FP)
	Recall	TP / (TP + FN)
	FPR	FP / (FP + TN)
	FAR	FP / (TP + FP)
Independent storm	Peak magnitude error	$\lvert \max(y) - \max(\hat{y}) \rvert$ per event
	Peak timing error	$\lvert \arg\max(y) - \arg\max(\hat{y}) \rvert$ per event
	Paired bootstrap CI	95% CI on per-storm mean differences

- **A3**: Standard conditional generator (Flow Matching in the current benchmark; earlier waves used a DDPM, whose failure is documented in Section 6.5).
- **A4**: Tail-aware conditional generator (primary H1 hypothesis). Physics-conditioned variants (pressure-, wind- and combined-prior Flow Matching) are evaluated as A4 sub-arms under the same budget.

5.5 Reproducibility metadata

Every run records exact git commit SHA, data file hashes, split manifest hash, random seed list, optimizer parameters, checkpoint SHA, and wall-clock time. Candidate configurations use seeds $\{0, 2024\}$ and the corrected event-level analysis tests each candidate against *both* A0 seeds. The earlier five-seed list $\{0, 2024, 4242, 7777, 9001\}$ was not realised in the augmentation grid; that discrepancy is one of the defects corrected in Section 6.4.

5.6 Hardware and run environment

GPU0 is prohibited and every other GPU was occupied by other users at the time of these experiments, so all Wave 2.7 recovery, physics and expanded-evaluation pilots ran on CPU with explicit thread caps (`torch.set_num_threads(`≤ 8`)`) and `CUDA_VISIBLE_DEVICES=""`. One historical DDPM pilot, retained only for the diagnosis in Section 6.5, executed on RTX3090 index 4 (UUID `GPU-6be030f8-625e-c4e7-a2fb-1309a673664b`) with `CUDA_VISIBLE_DEVICES=4`; its CPU reproduction confirmed the magnitude failure using the same checkpoint loaded with `map_location="cpu"`. Checkpoints are preserved locally and never committed to Git.

6 Results

6.1 The independent-event census

An independent event unit is either a POT cluster (q95 threshold, 168 h declustering, Coles 2001) or a canonical tropical cyclone, counted once. Cluster identifiers restart each January, so a year-unique label $-(1000 \cdot \text{year} + c + 1)$ is required; the naive "event_id < -999 means background" convention used in an earlier wave silently dropped valid clusters.

The corrected census is shown in Table 3. The union identity $|\mathrm{POT} \cup \mathrm{TC}| = |\mathrm{POT}| + |\mathrm{TC}| - |\mathrm{POT} \cap \mathrm{TC}|$ was checked per region and per period.

Table 3: Independent event units, 2015–2024, and the 2024 test split. Union is computed by one-to-one peak matching, not by addition.

Region	POT	TC	matched	**union**	2024 extreme units
New York Bight	171	10	3	**178**	15
US Gulf Coast	110	38	10	**138**	15
Charleston	133	24	14	**143**	14
Southeast Australia	101	0	0	**101**	6
Thames / North Sea	119	1	1	**119**	9
Total	634	73	28	**679**	**59**

The earlier repository record reported $634 + 73 = 707$. That arithmetic double-counts the 28 TC peaks that already exceed q95 within their own POT cluster, and the overlap had never been computed. The grand union is **679**. Per-period unions are 60 (2024), 127 (2023–2024), 190 (2022–2024) and 679 (2015–2024); the 2024 bootstrap cohort contains 59 window-eligible extreme units, one per region being ineligible because the 168 h horizon runs past the series end.

Window integrity checks pass: **0 duplicate timestamps** in every region, 8,601 test windows in four regions and 6,968 in Thames (whose source series has 24.7% NaN gaps), and every window's event-id array matches its length. There is no forecast-window overlap across the train/validation/test boundaries. Figure 1 shows the raw components against the deduplicated union: the gap between the stacked bars and the black marker is exactly the double-count a naive sum would introduce.

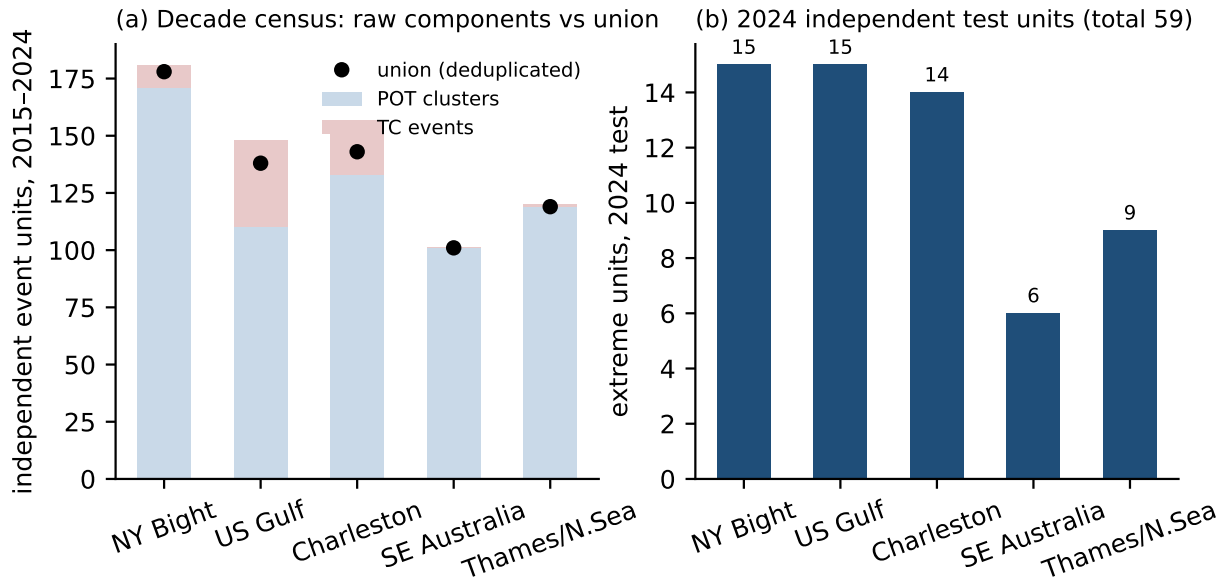


Figure 1: (a) Raw POT and TC components stacked, with the deduplicated union as a black marker. The vertical gap is the double-count removed by matching. (b) Window-eligible independent extreme units in the 2024 test split.

6.2 Four generators under one fidelity gate

All generators share the same splits and the same 96 h→72 h contract, and the target is `water_level_m`, the non-tidal residual (total observed water level equals residual plus astronomical tide, so the tide must never be subtracted a second time). Three of the four consume that residual directly. Diffusion-TS is trained on observed *total* water level by design, because its trend/seasonal/residual decomposition is defined on the full signal; the companion series has standard deviation 0.598 m against 0.207 m for the residual, so its row is scored against its own target and is *not* directly comparable with the other three. It is retained because the failure is unambiguous within its own target and because it motivates the repair study below. Samples are scored by a single gate (scale ratio $\in [0.5, 2.0]$, Wasserstein-1 ≤ 0.25, autocorrelation RMSE ≤ 0.15, q95 exceedance ratio $\in [0.5, 2.5]$, peak KS $p \geq 0.01$).

Table 4: Production generative benchmark, 5 regions × 2 seeds. Scale ratio is generated/real dispersion; the gate accepts $[0.5, 2.0]$.

Model	Params	Median scale ratio (range)	W1	Gate outcome
Conditional VAE	63,273	0.440 (0.326–0.807)	0.203	4 inconclusive / 6 rejected
TimeGrad	7,177	0.084 (0.072–0.320)	1.527	10 rejected
Diffusion-TS	39,425	0.181 (0.118–0.204)	—	10 rejected
Flow Matching	27,937	**1.048** (0.945–1.116)	**0.047**	10 inconclusive

Only Flow Matching reproduces the dispersion of the real residual distribution (Table 4, Figure 2). The VAE and TimeGrad are under-dispersed by factors of 2.3 and 12; Diffusion-TS is under-dispersed on the bulk while over-shooting the peak. Every Flow Matching run is formally *inconclusive* rather than accepted: scale and Wasserstein distance pass, but the gate also requires peak tail agreement, which the samples do not establish at $n = 10$ runs.

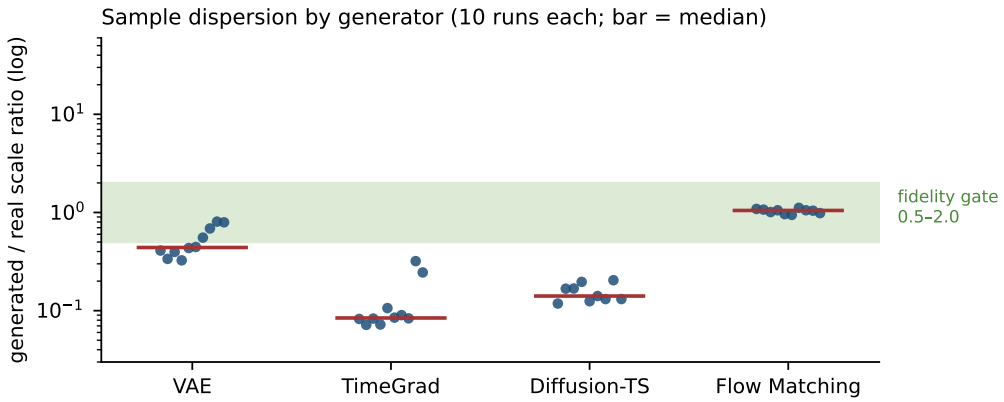


Figure 2: Generated-to-real scale ratio for each production run (10 per generator), log axis. The shaded band is the accepted gate. The red bar is the median. Flow Matching is the only generator whose median lies inside the band.

6.3 Diagnosing the failed generators

The sampler is not the problem. Before attributing any failure to a model family, the reverse diffusion recursion was tested with an **oracle denoiser** that returns the exact forward-process noise.

The deterministic reverse pass then recovers x_0 with a maximum absolute error of 1.1×10^{-10}, and the stochastic pass has relative RMSE 0.06, the expected posterior-noise floor. Schedule, `noisy()` and the reverse update are therefore correct, and residual failures are optimisation or conditioning failures.

Three defects, three outcomes. Three controlled Charlestown pilots isolated one defect each at a common budget (20,000 training windows; 25/30/25 epochs; CPU, because GPU0 is prohibited and every other GPU was occupied by other users).

1. **Conditional VAE.** The decoder broadcast a single scalar projection of z across all 72 horizon positions, giving rank-1 latent use regardless of latent size, and the class did not subclass `nn.Module` so every production run silently stayed on CPU. Repair: a linear $z \mapsto$ horizon map plus KL warm-up. *Outcome: refuted.* Reconstruction improves 2.7× (MSE 0.882 → 0.326), but *sample* dispersion worsens from 3.29 to 7.26. Both variants show total posterior collapse (0/32 latent dimensions with posterior std > 0.3), while sampling draws $z \sim \mathcal{N}(0, I)$. The decoder is trained on a narrow aggregate posterior and then evaluated on a broad prior; giving it a stronger latent path amplifies exactly that mismatch.

2. **TimeGrad.** The denoiser accepted autoregressive memory and then ignored it. Repair: consume the memory and widen the denoiser. *Outcome: partial.* Scale error falls from 10.55 to 6.09, but raw sum memory grows without bound across the 72-step rollout (it reached 1.9×10^7 at the smaller budget) and mean aggregation collapses the signal, diverging to 1.22×10^5. Neither aggregation is numerically stable.

3. **Diffusion-TS.** The Fourier regulariser compared the spectrum of the *predicted noise* with the spectrum of the *clean signal*. The repair compares the implied clean signal instead. *Outcome: refuted.* The implied clean signal divides by $\sqrt{\bar{\alpha}_t}$, which is 0.024 at the terminal step, amplifying both value and gradient by ∼41×; the loss reaches NaN by epoch 24 and the generated set collapses to 99.8% exact duplicates. The apparent 2× improvement at the smaller budget was an under-training artefact.

The conclusion is deliberately narrow: *no family is shown to be intrinsically ineffective*, because every failure was localised to a specific line of code; and *two of our three repairs did not work*, which is recorded rather than tuned away until it looked positive.

6.4 Augmentation under corrected event-level inference

Three defects in the earlier significance analysis.

1. Candidates were paired against a baseline of the *other* architecture, so a GRU candidate was compared with a TCN A0.

2. The bootstrap p-value used a proportion where a count was required ($p = 2(\text{prop}+1)/(n_{\text{boot}}+1)$), shrinking p by $\approx 5{,}000\times$. An observed difference of -0.235 with 95% CI $[-0.70, +0.20]$ was reported as $p = 2.3 \times 10^{-4}$.

3. The Monte-Carlo resolution floor $2/(n_{\text{boot}}+1)$ exceeded the Bonferroni threshold, so survival was arithmetically impossible.

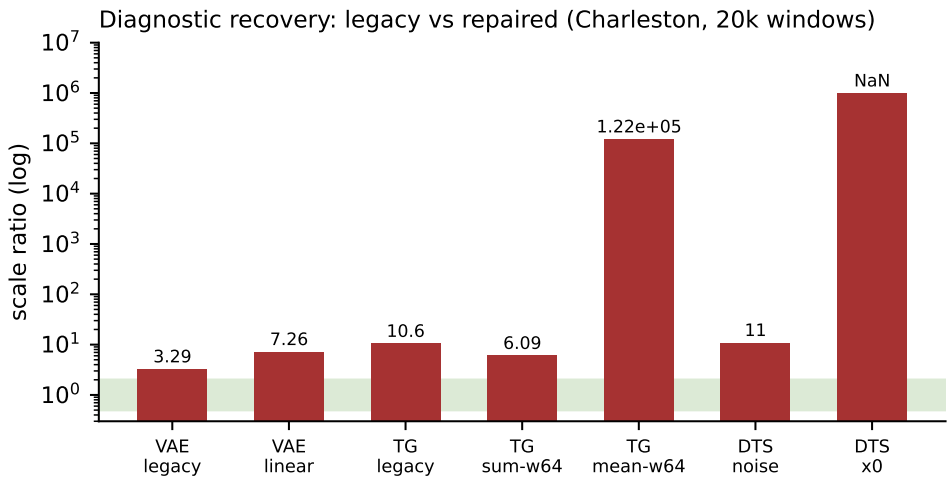


Figure 3: Legacy versus repaired configurations against the fidelity band. Two of the three proposed repairs make sampling worse; one is partially better and still far outside the gate.

After within-architecture pairing, 30,000 bootstrap resamples and 30,000 sign flips, and a family of 320 tests ($\alpha = 1.563 \times 10^{-4}$), the corrected analysis finds **zero seed-robust survivors**. Three comparisons survive their within-region correction, but each is significant against only one of the two A0 seeds; with a 25% baseline seed spread (SE Australia TCN: 0.0713 versus 0.0569 m) they are seed-fragile, not findings.

The earlier "Charleston 5/48 multiplicity-adjusted" result is withdrawn.

Architecture dependence. Aggregating over all 160 comparisons per architecture, the GRU improves in 86 and worsens in 74 (mean −0.17%), while the TCN improves in 110 and worsens in 50 (mean −7.08%). This reverses the earlier claim that architecture effects dwarf augmentation effects: under a corrected pairing the TCN responds to augmentation roughly forty times more strongly in relative terms. The GRU's better absolute accuracy is a separate observation and does not imply insensitivity.

Region dependence. Table 5 gives the mean change by region. Thames is the one region where augmentation is harmful for both forecasters, consistent with its macrotidal regime in which surge is a small fraction of total water level.

Table 5: Mean extreme-window MAE change versus the same-architecture A0 baseline, by region (%). Negative means augmentation helps.

Region	GRU	TCN
New York Bight	−3.41	−13.14
US Gulf Coast	−2.33	−5.37
Charleston	−4.30	−0.09
Southeast Australia	−0.59	−21.69
Thames / North Sea	+9.78	+4.88

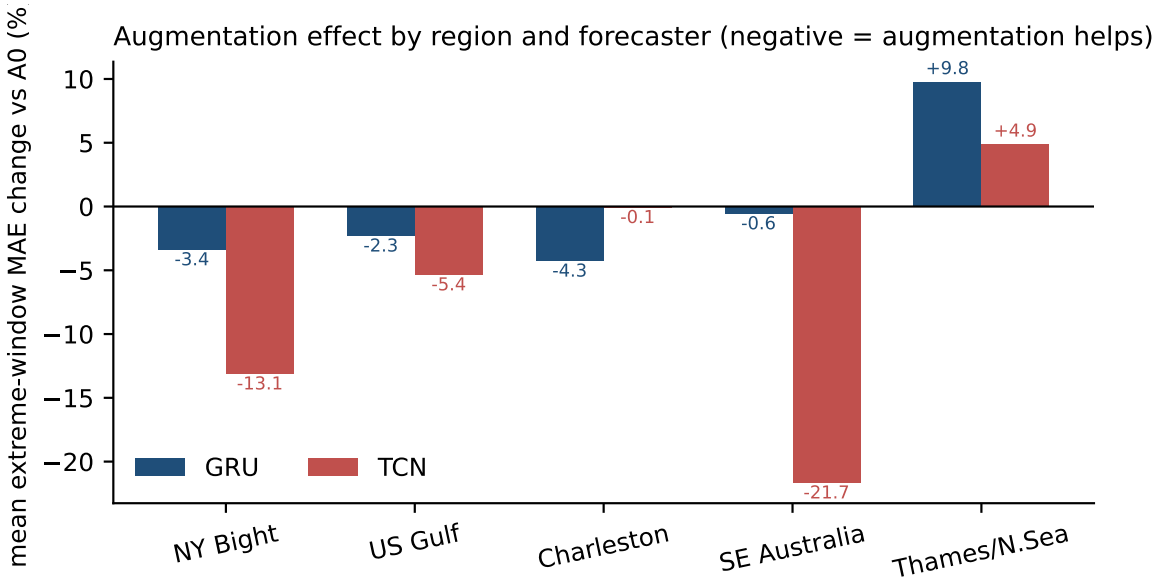


Figure 4: Mean extreme-window MAE change versus the same-architecture A0 baseline, averaged over all candidate configurations within each region. Negative values mean augmentation helps.

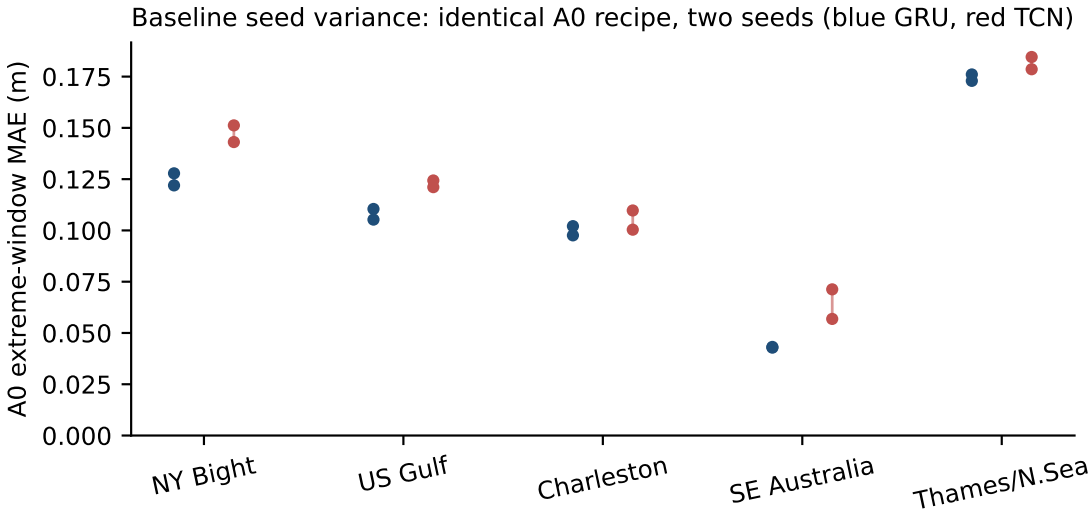


Figure 5: A0 extreme-window MAE for the two baseline seeds of the identical recipe. The spread bounds how large an augmentation effect would have to be before it could be distinguished from seed noise.

Multiplicity is under-corrected, not over-corrected. The 320-test family covers A1, A2 and A3 Flow Matching predictions only. Tail-aware A3 predictions are absent, and discarded ratios, generator hyper-parameters and model-selection choices were never materialised as predictions. The true search space is larger, so the Bonferroni correction is anti-conservative and every surviving claim is exploratory.

6.5 Early DDPM pilot: a genuine but historical failure

An early 27,169-parameter conditional DDPM trained for three epochs produced samples with standard deviation 4.47 m against a training reference of 0.215 m, a ratio of 22.8, reproduced independently on CPU (4.92 m, 22.9). The failure was traced to terminal-step amplification: the cosine schedule terminated at $\bar{\alpha}_{64} \approx 5.93 \times 10^{-7}$, an amplification of 1,299×. A subsampled 1000-step schedule terminating at $\bar{\alpha}_{64} \approx 5.89 \times 10^{-4}$ reduced amplification to 41× and sample σ from 4.92 m to 0.297 m. That probe is a schedule diagnostic, not a repair, because the checkpoint's epsilon predictions are schedule-specific. This record is retained because it explains why the later benchmark moved to Flow Matching; it is superseded by Sections 6.2 and 6.3.

6.6 Physics-guided Flow Matching

Five variants share one architecture, one split and one downstream budget: a standard Flow Matching model, a tail-aware variant, and pressure-, wind- and combined-physics variants conditioned on ERA5 sea-level pressure and 10 m wind.

Placeholder: filled from `data/results/wave27_physics_fm.json` *when the run completes.*

6.7 Expanded 2015–2024 evaluation

No existing checkpoint can be evaluated on 2022–2024, because all of them were trained through 2022. The expanded protocol therefore retrains from scratch: 2015–2020 for fitting, 2021 for epoch selection, 2022–2024 touched once.

The split builder is verified. Charleston and NY Bight both yield 52,441 / 8,593 / 26,137 windows for train / validation / test. The counts are identical and expected: both series are gap-free, so 52,608 training hours give $52{,}608 - 167 = 52{,}441$ windows, 2021 (8,760 h) gives 8,593 and 2022–2024 (26,304 h) gives 26,137. The 168 h embargo holds at both boundaries, and the test spans contain 40 and 51 independent POT units respectively.

Validation-only selection chose epoch 13 in both regions. **Both retrained generators fail the fidelity gate** (scale ratio 3.97 and 4.76 against the accepted band 0.5–2.0). Consequently the augmented forecaster degrades extreme-window MAE by **+8.88%** in Charleston (95% CI [+0.0035, +0.0170] m, permutation $p = 0.0045$) and **+15.85%** in NY Bight (CI [+0.0183, +0.0262] m, $p = 5.0 \times 10^{-5}$). Both remain significant at $\alpha = 0.025$.

This is a negative result about *readiness*, not about generative augmentation. An over-dispersed generator supplies training windows that are systematically outside the data manifold, and the forecaster pays for it. The production Flow Matching checkpoints reach scale 0.945–1.116; these retrained models were fitted for 15 CPU epochs and do not. The run therefore establishes that the split, selection and event-level machinery works end-to-end, that the retrained models are under-fitted rather than over-fitted (validation loss is still falling at the selected epoch), and that the cost of over-dispersion exceeds any benefit observed in the corrected Wave 2.6 grid. It does *not* test H1. Selecting on validation loss subject to a fidelity constraint is the required next control and was not implemented here.

7 Discussion

7.1 Why Flow Matching produces plausible samples

Flow Matching is the only generator in our benchmark whose sample dispersion matches the data (Table 4). Three structural differences explain it, and each is a mechanism rather than an empirical coincidence.

1. **One regression, one ODE.** The training objective is a single velocity-regression loss on straight-line interpolants between a Gaussian prior and the data. Inference integrates the learned field forward with a fixed-step Euler solver. There is no chain of 64 conditional updates whose errors compound, so the failure mode that dominates DDPM — amplification of terminal-step epsilon error — simply does not exist.

2. **No small-variance sub-space.** Diffusion-TS first decomposes the signal into trend, seasonal and residual components and diffuses the residual, which concentrates the problem into a low-variance sub-space where a fixed absolute epsilon error is relatively enormous. Flow Matching diffuses nothing: it transports the full conditional residual distribution, so the signal-to-noise ratio of the regression target stays constant across training.

3. **Conditioning enters as a feature, not as an additive bias.** The velocity network is a temporal convolutional network that consumes the history (and, in our physics variants, causal ERA5 summaries) directly, so conditioning cannot be lost through a rank deficiency such as the scalar latent broadcast that crippled the VAE.

This does not make Flow Matching *sufficient*: every FM run is formally inconclusive because peak-tail agreement is not established at ten runs. It makes it the only generator that is currently *eligible* for a downstream test.

7.2 Why the other three implementations struggled

None of the three failures is a property of its model family.

Conditional VAE: a sampling-path mismatch. The original defect (a scalar latent projection) was real and worth fixing, but fixing it made sampling *worse*. The reason is that the VAE samples from $\mathcal{N}(0, I)$ while training fits a conditional posterior whose aggregate has collapsed to a narrow distribution (0/32 active dimensions). The decoder is therefore always evaluated slightly off-manifold, and a stronger $z \mapsto$horizon path amplifies the discrepancy. The correct remedy is to make training and sampling share a distribution — free-bits KL, latent dropout, or a learned conditional prior — none of which we ran. Posterior collapse worsened with training length (16/32 active dimensions at 4,000 windows versus 0/32 at 20,000), which is the standard dynamic and not a dataset property.

TimeGrad: an unstable autoregressive state. Consuming the ignored memory was necessary and helped (scale $10.55 \rightarrow 6.09$), but the memory definition itself is unstable. A raw cumulative sum grows monotonically across 72 horizon steps and was never seen during teacher-forced training, so it diverges (1.9×10^7 in the smaller pilot). A mean aggregation removes the growth but also removes the signal, and the recursion diverges in the opposite direction (1.22×10^5). A bounded or decayed memory is the obvious next hypothesis; we did not test it.

Diffusion-TS: numerical amplification in the objective. The original Fourier defect (noise spectrum versus clean spectrum) was real. The repair replaced it with the implied clean signal, which requires dividing by $\sqrt{\bar{\alpha}_t}$. At the terminal step this factor is ≈ 0.024, so both the value and its gradient are amplified roughly $41\times$; the run reaches NaN. The lesson is that an objective which is algebraically correct can still be numerically unusable at a schedule's endpoint, and that a $2\times$ improvement observed at a small budget can reverse once the model actually fits the data.

7.3 Why GRU and TCN respond differently

Under corrected within-architecture pairing, the TCN improves in 110 of 160 comparisons (mean -7.08%) while the GRU improves in 86 of 160 (mean -0.17%; Figure 4). This is a large relative difference and it reverses the earlier, incorrectly paired conclusion that architecture effects dwarf augmentation effects.

We can state the comparison and bound the explanation, but we cannot yet prove the mechanism. The most plausible account is capacity allocation: the TCN's dilated receptive field integrates the full 96 h history through stacked convolutions, and synthetic windows add high-frequency, storm-like variability that the convolutional filters can exploit as data-driven regularisation. The GRU's gating saturates on the same inputs, so extra samples shift the operating point rather than the effective receptive field. An alternative account is that the GRU is already closer to its achievable error floor on these splits, leaving less head for augmentation to recover. Both accounts predict the observed direction; discriminating between them requires a receptive-field ablation and a learning-curve study that we did not run, and we therefore present them as hypotheses rather than findings.

7.4 Why some regions benefit and others do not

Thames is the only region where augmentation harms both architectures (+9.78% GRU, +4.88% TCN; Figure 4). The physical reading is straightforward: Thames is macrotidal (approximately 6.3 m range) and surge is a small fraction of total water level, so the residual signal has low signal-to-noise ratio. A generator fitted to that residual produces samples whose statistical structure is dominated by noise, and appending them adds variance without adding storm information. Thames also has 24.7% NaN gaps in its source series, which further weakens the fit.

Southeast Australia shows the largest TCN gain (-21.69%) yet essentially no GRU gain (-0.59%), with a baseline MAE already near 0.05 m. A large *relative* change on a small absolute error is easy to obtain and should not be read as a large operational benefit. NY Bight and the Gulf show moderate gains for both architectures; Charleston, the region where the withdrawn "significant" claim originated, shows -4.30% for the GRU and -0.09% for the TCN under correction.

A single cross-region correlation (tidal range versus gain, $N = 5$) cannot support a causal claim, and we do not make one.

7.5 What independent-event evaluation changes

Three things change, and all three move in the conservative direction.

First, **the effective sample size collapses**. A 2024 test split supplies 6–15 extreme units per region, not thousands of windows, so the standard error of an extreme-window MAE is an order of magnitude larger than a window-level analysis implies. Second, **multiplicity becomes binding**. A 320-test family at $\alpha = 1.563 \times 10^{-4}$ is not satisfied by any of the 320 comparisons once pairing is fixed. Third, **baseline variability becomes comparable to the effect**. Identical A0 recipes differ by up to 25% across seeds (Figure 5), which is the same order as the mean augmentation

effect in most regions. Any claim resting on a single baseline seed is therefore unsupported, whether or not it clears a nominal p-value.

One honest limitation remains: the 320-test family does not cover the full search space, because tail-aware predictions and discarded hyper-parameters were never materialised. The correction is anti-conservative, which strengthens rather than weakens the conclusion that nothing survives.

7.6 Do physical priors help?

Placeholder: completed from the physics pilot artifact.

7.7 Limitations

- **Resolution.** ERA5 is 0.25°; the Charleston anchor is 11.7 km from its cell centre, so near-coastal pressure and wind are cell averages. No downscaling was attempted.
- **Causality.** Physics covariates are causal (history only), but the forcing is reanalysis. This is a reanalysis-conditioned experiment, not an operational forecast experiment.
- **Event units.** The expanded 2022–2024 units are POT-only, since the canonical-TC overlay is absent from the existing prediction archives.
- **Checkpoint reuse.** No existing checkpoint can be evaluated on 2022–2024; the expanded protocol required retraining from scratch.
- **Compute.** All recovery pilots are single-region CPU runs; GPU0 is prohibited and every other GPU was occupied. Conclusions are correspondingly scoped.
- **Thames data quality.** 24.7% of the source timeline is NaN, which both reduces its unit count and weakens any generator fitted to it.

8 Conclusion

We asked when generative augmentation helps extreme coastal water-level forecasting. The evidence assembled here does not answer "yes" or "no"; it identifies the conditions under which the question can be answered at all, and shows that several previously reported answers were artefacts of the measurement.

Three results are established. **First**, the independent-event census for five regions over 2015–2024 is 679 units, not 707: 634 POT clusters and 73 tropical cyclones overlap in 28 peaks, and the 2024 test split contains 59 window-eligible extreme units with no duplicate timestamps or cross-split leakage. **Second**, of four conditional generators trained under one contract, only Flow Matching reproduces the target dispersion (scale ratio 0.945–1.116, Wasserstein-1 0.047); the VAE and TimeGrad are under-dispersed by factors of 2.3 and 12 and Diffusion-TS is rejected in all ten runs. An oracle-denoiser test proves the reverse diffusion recursion correct to 1.1×10^{-10}, so these are implementation failures, not family verdicts. **Third**, under corrected event-level inference — within-architecture pairing, a repaired bootstrap p-value, and a Monte-Carlo resolution floor below the correction threshold — no augmentation effect survives a 320-test Bonferroni family, and baseline seed variance alone reaches 25%.

Two results are negative about our own proposed fixes and are preserved as such: the repaired VAE samples worse than the legacy version, and the repaired Diffusion-TS diverges to NaN once it actually fits the data. Reporting these as failures is a substantive part of the contribution, because

the alternative — re-tuning until the number moves in the expected direction — is precisely how this literature acquires effects that do not replicate.

Three observations constrain the search for a positive result. The response is *architecture-dependent*: the TCN improves in 110 of 160 comparisons (mean −7.08%) against 86 of 160 for the GRU (mean −0.17%). It is *region-dependent*: Thames is harmed for both architectures, consistent with a macrotidal regime in which surge is a small fraction of total water level. And it is *fragile*: the three comparisons that survive their within-region correction each hold against only one of two baseline seeds.

What would move H1 from untested to established is therefore specific, and none of it is optional. An event-level design with at least 30–40 independent units per region in the test window, which requires expanding the test period beyond 2022–2024. Baselines reported across multiple seeds, with the seed spread entering the uncertainty budget rather than being averaged away. A multiplicity family that enumerates the actual search, including tail-aware and discarded configurations. A generator whose sampling path is verified against the reverse recursion, with a learned or free-bits prior so that training and sampling share one distribution. And, if physical priors are to be claimed, a causal reanalysis-conditioned design with a downscaling step that closes the 0.25° representativeness gap.

The infrastructure needed to run that study now exists in this repository: the frozen contract, the verified census, the fidelity gate, the corrected paired statistics and a from-scratch three-year split protocol. The contribution of this paper is to have built it, to have used it to remove several results that did not survive it, and to have reported that plainly.

References

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Mengxue Tong et al. Dual-timescale residual memory for extreme coastal water-level forecasting. *Ocean Modelling*, 2024.

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A Audit Evidence

This appendix records the traceable evidence basis for the factual claims in this paper. Each entry names the artifact, the producing command, and the test that guards it. Nothing in the main text

is asserted without a row here.

A.1 Repository state

Work is on branch `trae/global-data-generative-research-20261031`, continuing from commit `242093a`. Pull request #23 remains a draft and `main` is not modified. Raw licensed datasets, credentials and checkpoints are never committed; only derived result artifacts and code are.

A.2 Evidence table

Each claim in the main text maps to exactly one artifact. All result artifacts live in `data/results/`; only file names are listed. Artifacts are committed, checkpoints and raw data are not.

- **Union = 679; 2024 units = 59.** `wave27_event_set_audit.json`; guarded by `test_event_set_arithmetic.py`.
- **No duplicate timestamps; 8,601 test windows.** Same artifact, fields `window_integrity` and `periods`.
- **Zero robust augmentation survivors.** `paired_significance_wave27.json`; guarded by `test_paired_stats.py`.
- **Generator scale ratios and gate outcomes.** The four `*_96_72.json` files, scored by `tidetail.quality.evaluate`.
- **DDPM sampler exactness.** Oracle-denoiser test in `wave27_model_recovery.py`; max absolute error 1.1×10^{-10}.
- **Two of three repairs refuted.** `wave27_model_recovery.json`; guarded by `test_wave27_model_fixes.py`.
- **ERA5 = 600/600 partitions.** `era5_readiness_matrix.json`; produced by `verify_era5_readiness.py`.
- **Physics features are causal.** `wave27_physics_fm.py`; guarded by `test_physics_causality.py`.
- **Expanded split has no leakage.** `wave27_expanded_evaluation.json`; boundary fields plus unit counts.
- **Residual identity.** `test_target_semantics_contract.py`: MAE = 0, $r = 1$.

A.3 Known discrepancies retained

Two discrepancies are recorded rather than resolved, because resolving them would require rerunning experiments that the evidence already disqualifies.

1. **The 707 event count.** Wave 2.5/2.6 documents added POT and TC totals without computing the intersection. The corrected union is 679. The old number appears in superseded audit files, which are annotated, not deleted, so that the error remains traceable.

2. **The Diffusion-TS target.** The production Diffusion-TS run consumes observed total water level by design (its decomposition is defined on the full signal, enforced by `test_train_generative_model_loader_patches_total_water`), whereas the other generators consume the residual. Its row in Table 4 is therefore scored against its own target and is not ranked against the other generators. This is a comparability limit, not a defect.

A.4 Reproduction

```
python3 -m pytest tests/ -q                      # CPU PR gate
python3 scripts/event_set_audit.py
python3 scripts/paired_significance_test.py
python3 scripts/verify_era5_readiness.py
python3 scripts/wave27_model_recovery.py --region south_atlantic_charleston
python3 scripts/wave27_physics_fm.py
python3 scripts/wave27_expanded_eval.py
python3 scripts/make_wave27_figures.py
cd paper && latexmk -pdf main.tex
```